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Group Number

# RPEasy: A Rate of Perceived Exertion Estimation Tool for Athletes

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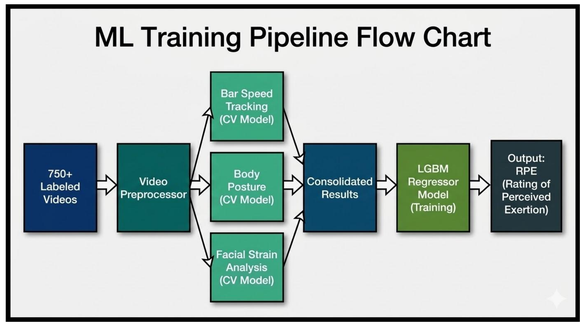
## INTRODUCTION

### Motivation & Objectives

- **Rate of Perceived Exertion (RPE)** -> subjective metric for determining lifter's training intensity
- Reliable **AI tool** that estimates an athletes RPE on a 1-10 scale




### Key Challenges:

- Video Noise, Dataset, Angles and Equipment

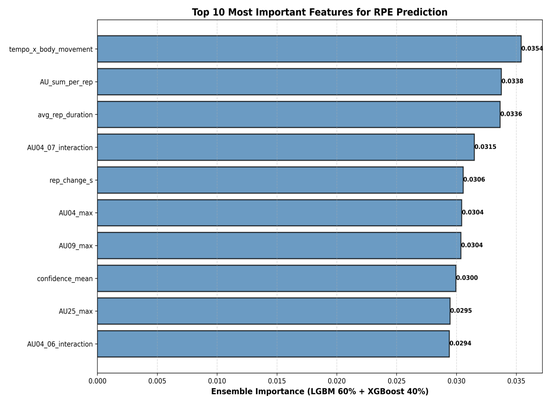
## ANALYSIS

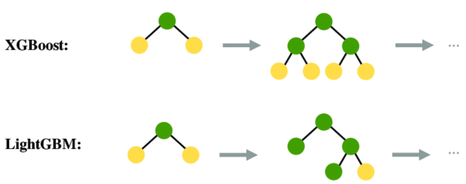
### Research:

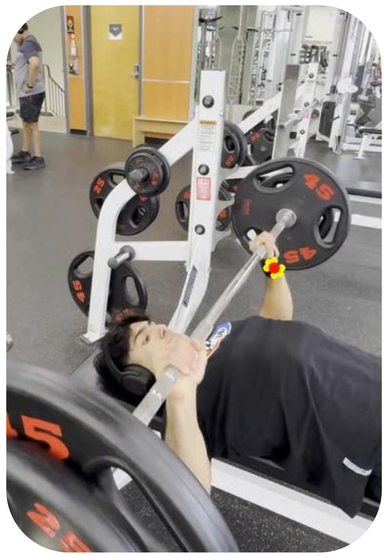
- RPE helps athletes understand if they are **over/under training**
- Key features can be used to **estimate RPE**

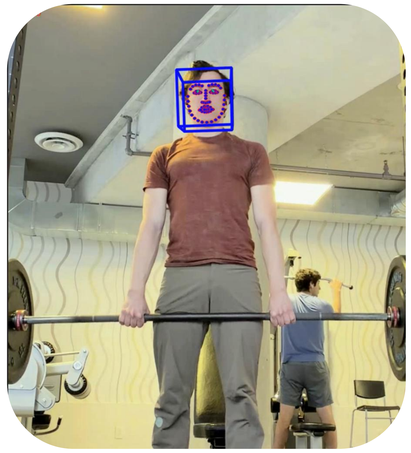
### What this means:

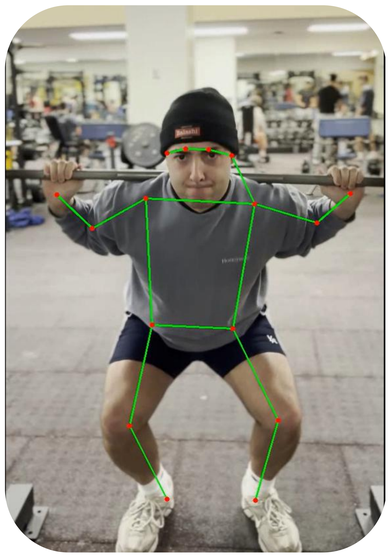
- **Machine learning** and **computer vision** to estimate RPE for athletes
- Bar path velocity, change in form, and facial strain are useful metrics

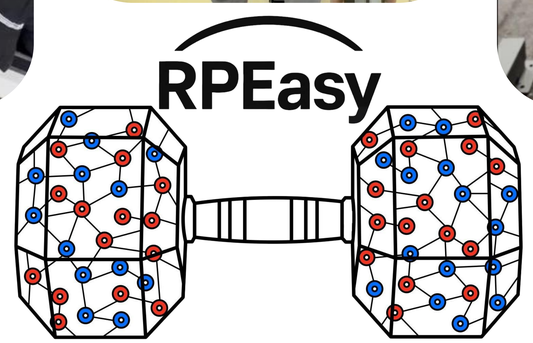


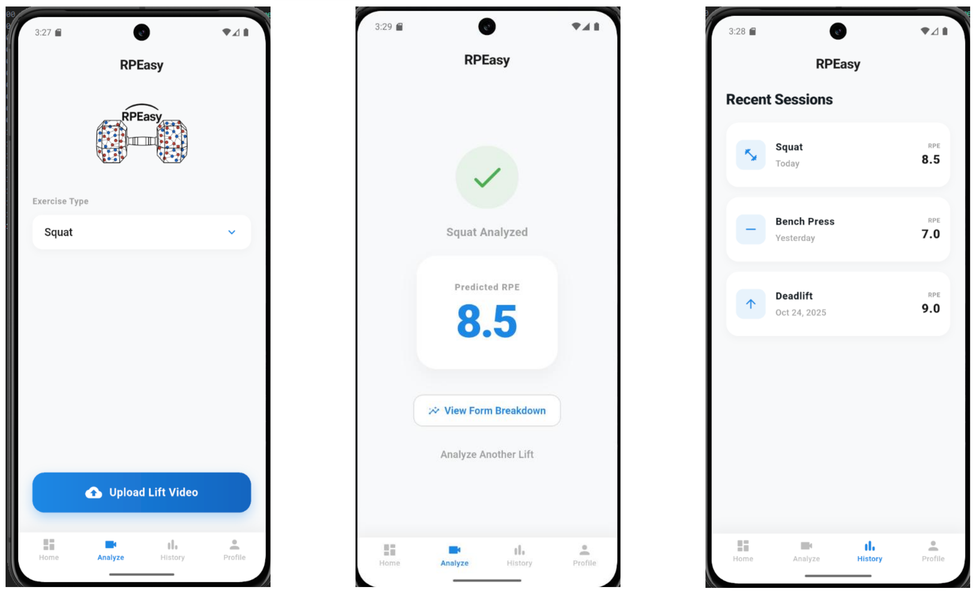







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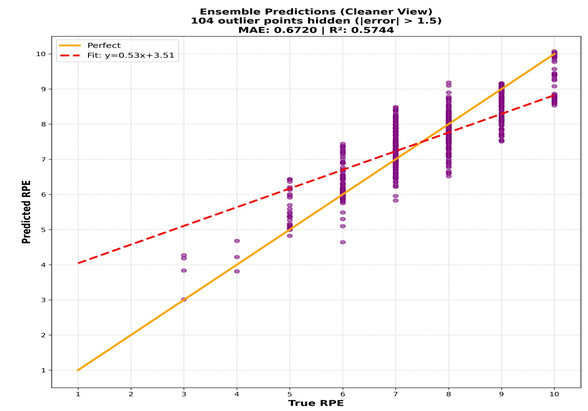
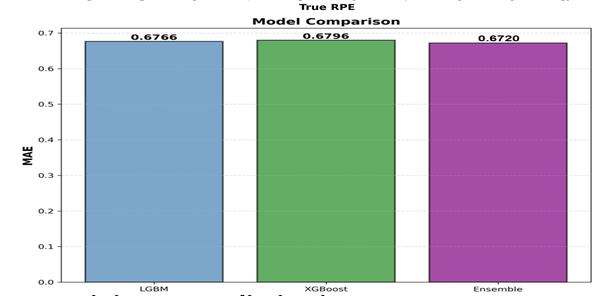
## CONCLUSIONS & RECOMMENDATIONS

### Conclusions:

- Combining bar path velocity, change in form, and facial strain can objectively estimate a lifter's RPE
- Achieved an RPE estimation **MAE of $\pm 0.67$**
- Offers athletes an objective tool to enhance training efficiency and reduce injury risk




### Remaining Issues/limitations:

- Estimation accuracy + confidence
- Environmental factors
- Video input quality

### Future Recommendations:

- Broaden training dataset
- Certified training advice and workout analytics

## REFERENCES

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